# NITHESH RAMACHANDRAN

AI & FULL-STACK DEVELOPER · B.TECH AI & DATA SCIENCE

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## SUMMARY

Second-year AI & Data Science undergraduate who ships applied-AI products end to end: React and Next.js front-ends, FastAPI and Node back-ends, PyTorch and scikit-learn models, delivered to the web, Android and the desktop. Built Qurious for Smart India Hackathon 2026 and DeepGuard, an explainable AI-face detector. Seeking a Summer 2027 software or AI engineering internship.

## EDUCATION

**B.Tech, Artificial Intelligence and Data Science** — Coimbatore Institute of Engineering and Technology *Expected May 2029*

## PROJECTS

**Qurious** · Smart India Hackathon 2026 *2026*
React · TypeScript · Three.js · Capacitor · Vitest · qurious-eight.vercel.app · github.com/Nithesh-cod/qurious

- Built an in-browser quantum computing lab for problem statement SIH26140 with Team X FORCE: a 12-qubit statevector simulator with two-way circuit ↔ code sync for OpenQASM 3, Qiskit, Cirq and PennyLane.
- Verified against Qiskit Aer on 22 circuits (max deviation $6.66 \times 10^{-16}$); 448 Vitest tests assert every physics claim in the lessons.
- Self-checking AI tutor runs its answers on the simulator and drops any that disagree; 1,090 offline topics; ships as an Android APK.

**DeepGuard** · Explainable AI-generated face detection *2026*
Python · PyTorch · FastAPI · React · Capacitor · github.com/Nithesh-cod/Deep-Guard

- Trained and evaluated two detectors: an EfficientNet-B0 + Grad-CAM baseline (ROC AUC 0.991 in-distribution) and a CLIP ViT-L/14 linear probe (ROC AUC 0.919 on a frozen 4,241-image out-of-distribution test).
- One FastAPI engine serves a React web app, an Android app, a Telegram bot and signed, verifiable PDF reports.

**TestAI** · AI test-automation platform *2026*
React · FastAPI · Supabase · GPT-4o · Playwright · scikit-learn · testai-orcin.vercel.app · github.com/Nithesh-cod/testai

- GPT-4o turns plain-language requirements into runnable Playwright, Cypress and Jest tests; human-approved self-healing selectors.
- Statistical flaky-test scoring and a gradient-boosting failure predictor; Supabase Postgres with auth and row-level security.

**Ramji Electronics** · Offline-first Android inventory & billing *2026*
Kotlin · Jetpack Compose · Room · MVVM · github.com/Nithesh-cod/ramji-electronics

- Voice commands in English, Hindi and Tamil become stock and billing actions via intent parsing and fuzzy product matching, fully offline; transactional billing, hashed PIN lock, PDF bill sharing and CSV export.

**TechFest Symposium** · College event platform *2026*
React · Vite · Node.js · Express · Firebase · github.com/Nithesh-cod/techfest-symposium

- Registration, QR entry passes scanned in the browser, auto-generated PDF certificates and Razorpay payments; Express API hardened with JWT auth, Helmet and rate limiting.

**Also built: CropAI** (crop / weed segmentation, live) · **FastBite** (canteen pre-orders, live) · **VoxelSpeak** (speech → 3D model in webcam AR) · **Stock Price Prediction** (Random Forest, LSTM) · **3D portfolio** (Next.js, React Three Fiber).

## EXPERIENCE

**Front-End Web Development Intern** · CodeAlpha (Remote) *Nov 2025 – Dec 2025*

- Built four HTML, CSS and JavaScript projects (image gallery, calculator, portfolio, music player), each reviewed against the spec.
- Refactored static markup into reusable components to cut duplication; shipped every task through Git and GitHub.

## ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2026** — built Qurious with Team X FORCE for problem statement SIH26140 (Smart Education).
- **12 Infosys Springboard certifications (2025–26)** — Applied Generative AI, AI-first Software Engineering, Prompt Engineering, OpenAI GPT Models, GPT-3 for Developers, Front-End Web Developer, HTML5, CSS3, JavaScript, and three Python courses.

## SKILLS

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| **Languages** | TypeScript, JavaScript, Python, Kotlin, SQL, HTML / CSS |
| **Front-end** | React, Next.js, Vite, Tailwind CSS, Three.js / React Three Fiber, Framer Motion, GSAP |
| **Back-end & data** | FastAPI, Node.js, Express, REST, JWT, Supabase (PostgreSQL), Firebase Firestore, Room |
| **AI / ML** | PyTorch, scikit-learn, TensorFlow, GPT-4o and other LLM APIs, Whisper, CLIP, prompt engineering |
| **Mobile & tooling** | Jetpack Compose, Capacitor, Playwright, Vitest, Git / GitHub, Vercel, Netlify |
| **Languages spoken** | Tamil (native), English (professional working) |